



Aerospace Environment

Module 2: The Atmosphere

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[Comments on the Quizzes \(or mini-exams\)](#)



Goals of This Module

- ❖ **Understand basic physics of the Earth's atmosphere**
 - ❖ Atmosphere composition, layers, regions, density
 - ❖ Atmospheric dynamics (variations), winds, tides, and waves
 - ❖ Atmospheric response to geomagnetic activity
- ❖ **Understand effects of the Atmosphere on Spacecraft operations**
 - ❖ Satellite drag
 - ❖ Oxygen effects
 - ❖ Other effects: erosion and sputtering
- ❖ **Learn about Atmospheric space missions, including CHAMP and GRACE**

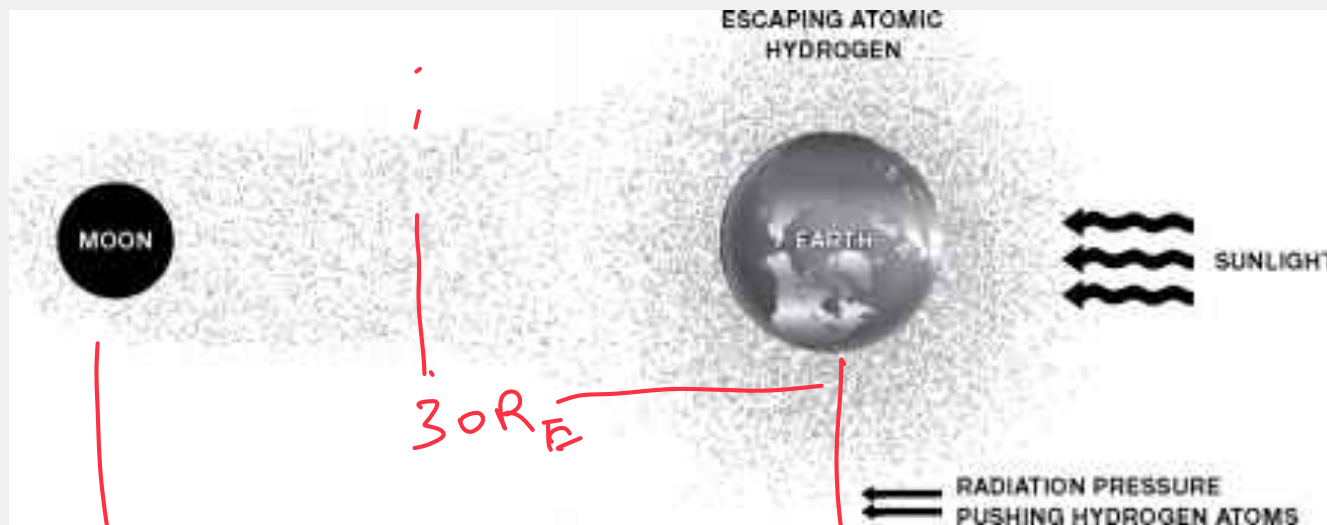
Geocorona

- ❖ Earth's atmosphere is 'visible' out to at least 15 Earth radii ($15 R_E$)
- ❖ Scattering of solar Lyman- α (121.4 nm) from hydrogen in the exosphere
- ❖ exosphere theoretically extends to $30 R_E$, where solar radiation pressure exceeds Earth's gravity
- ❖ where is the outer boundary of Earth's atmosphere (where the pressure of SRP 'balances' Earth's gravity)?

size: 14 $a_0 = 5.4 \times 10^{-11} \text{ m}$
 $m = 1.67 \times 10^{-27} \text{ kg}$

$SRP \propto (\pi a_0^2)$
 $\sim 10^{-26} \text{ N}$

$SRP = \frac{F}{c} = \frac{1361 \text{ W/m}^2}{c} \sim 4.5 \times 10^{-6} \text{ N/m}^2$



$F = mg_0 \left(\frac{R_E}{R_E + 30 R_E} \right)^2 = mg_0 \left(\frac{1}{31} \right)^2$
 $\sim 10^{-29} \text{ N}$

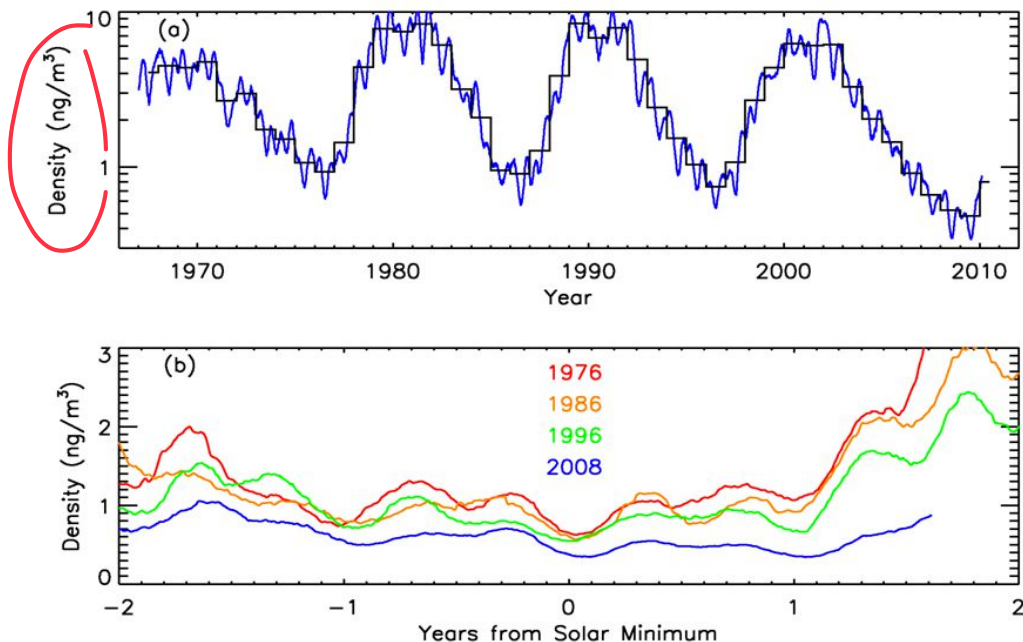
Atmospheric Variability

Solar-Cycle Variation

- ❖ Increase in solar EUV increases heating in the atmosphere, especially thermosphere;
- ❖ Increased heat => increased ionization (next module!)
- ❖ Increase in temperature => atmosphere rises, higher thermosphere density at a given altitude

order of magnitude

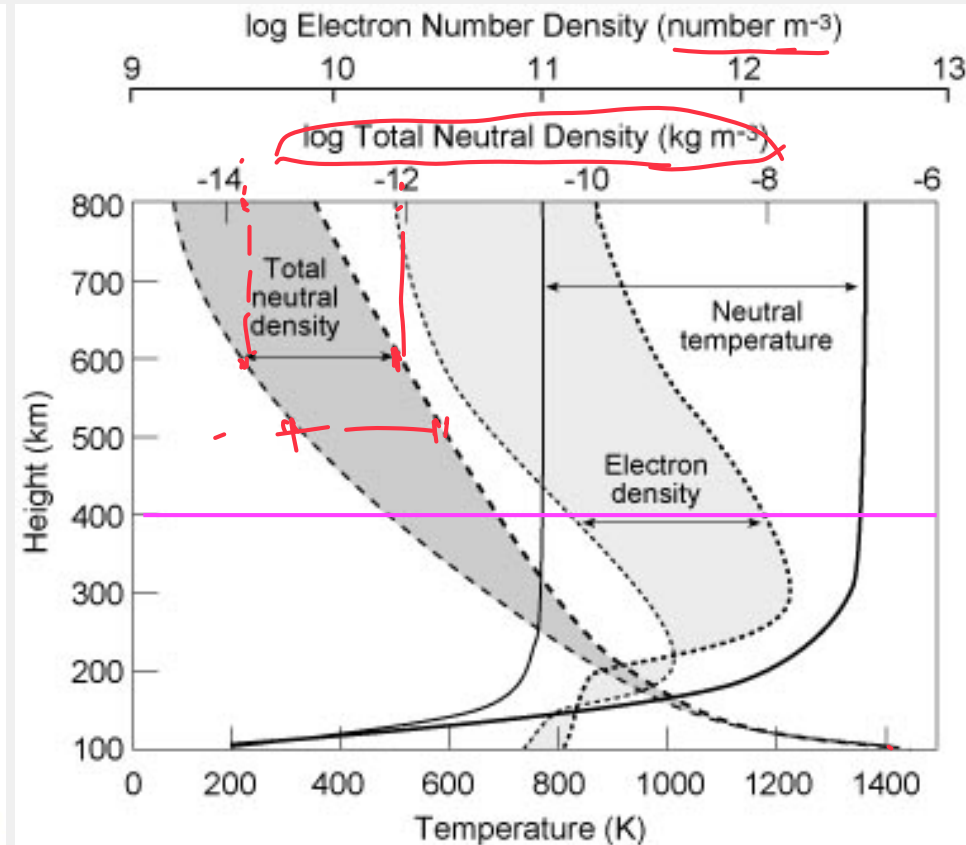
Long-Term Satellite Drag Data @ 400km



Global average neutral density at 400 km, 81-day average and annual average (top), and superposed epoch analysis (bottom)

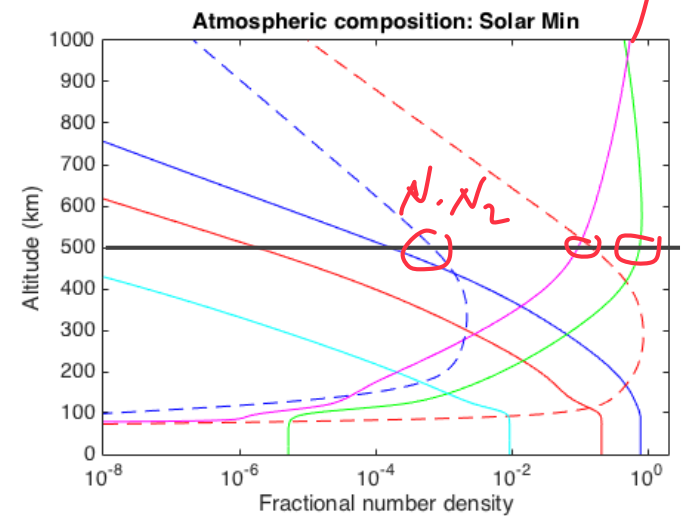
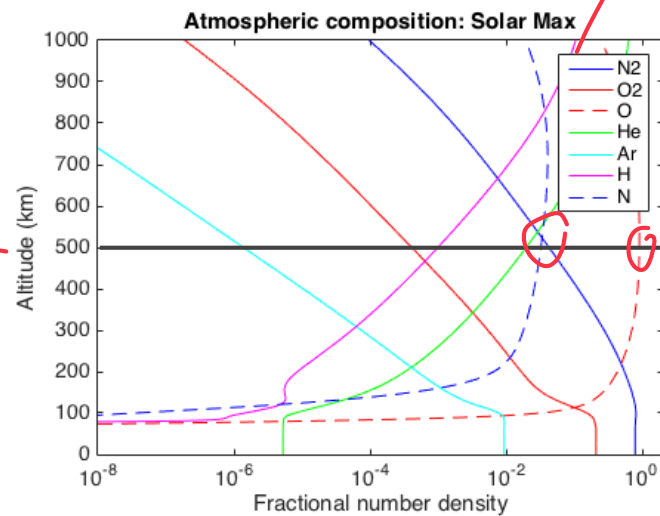
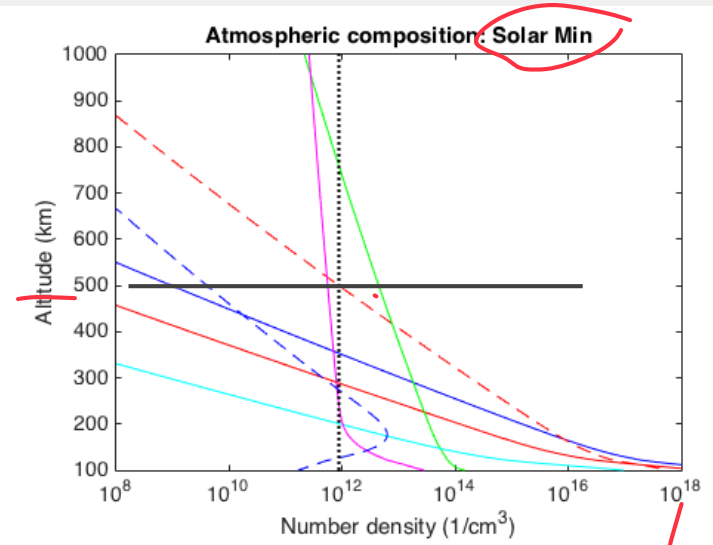
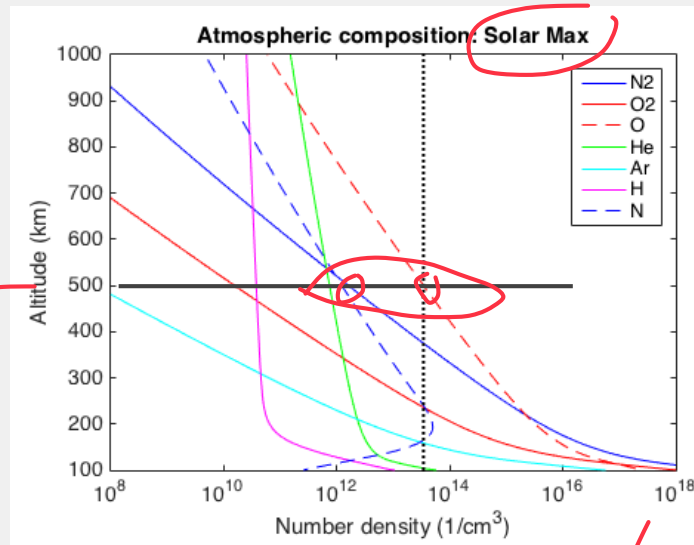
Emmert et al., *Geophys. Res. Lett.*, 37, L12102, 2010

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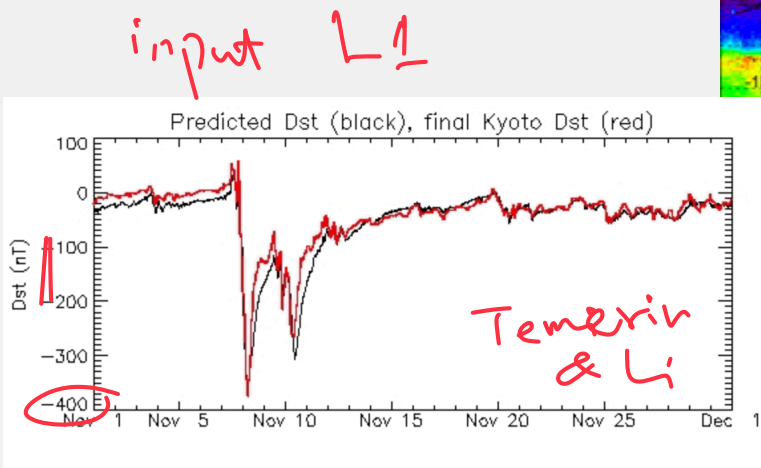
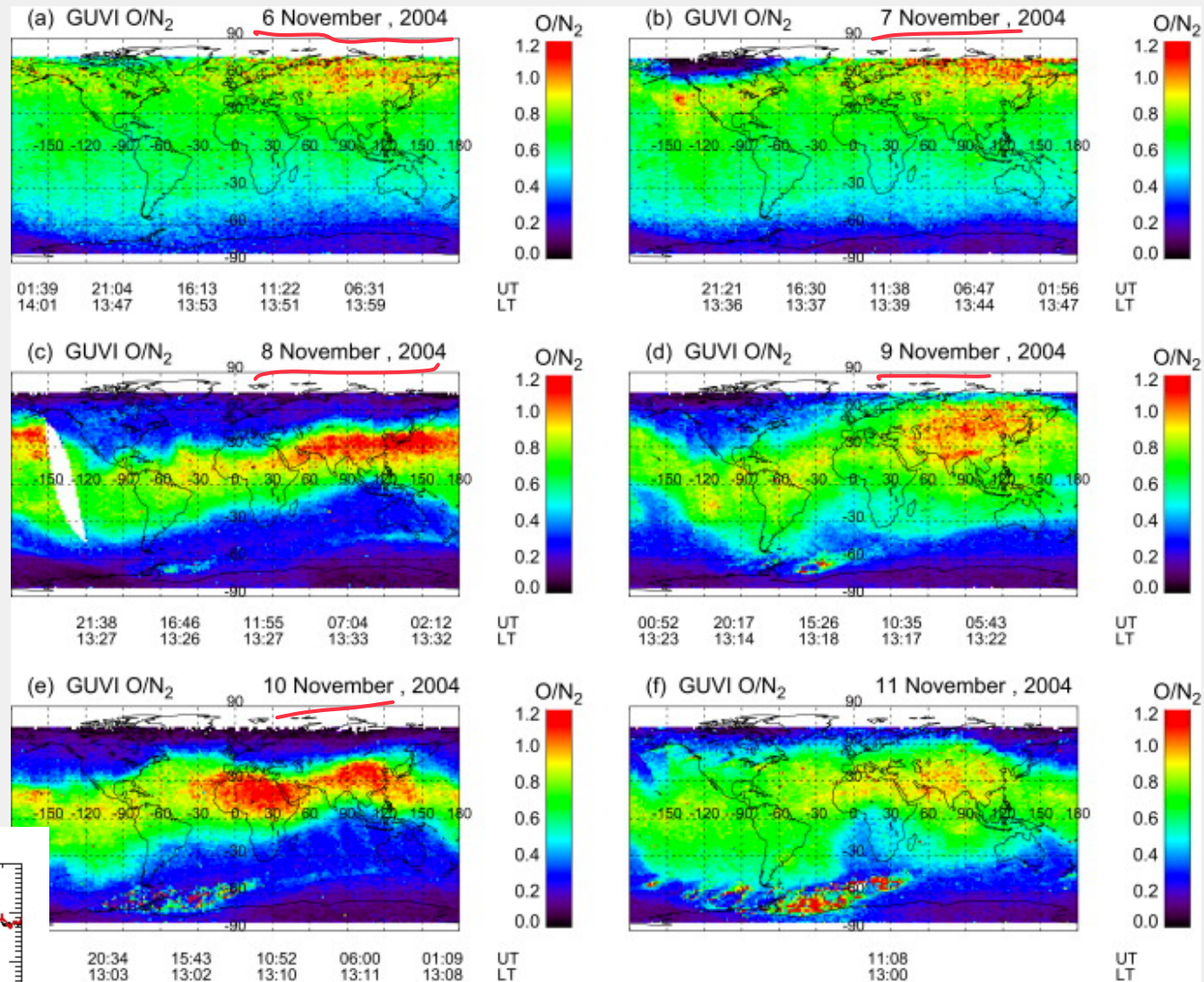
Composition Variation

- ❖ Change in temperature in the thermosphere (from 500 to 1500 K) affects diffusive equilibrium of components
- ❖ Generally, all particles get pushed to higher altitudes by higher temperatures below



GUVI (Global UV Imager) O/N₂ ratio

- ❖ GUVI (on TIMED satellite) measured the column density ratio of O to N₂; used as a proxy for neutral composition / heating of the atmosphere



real time forecast

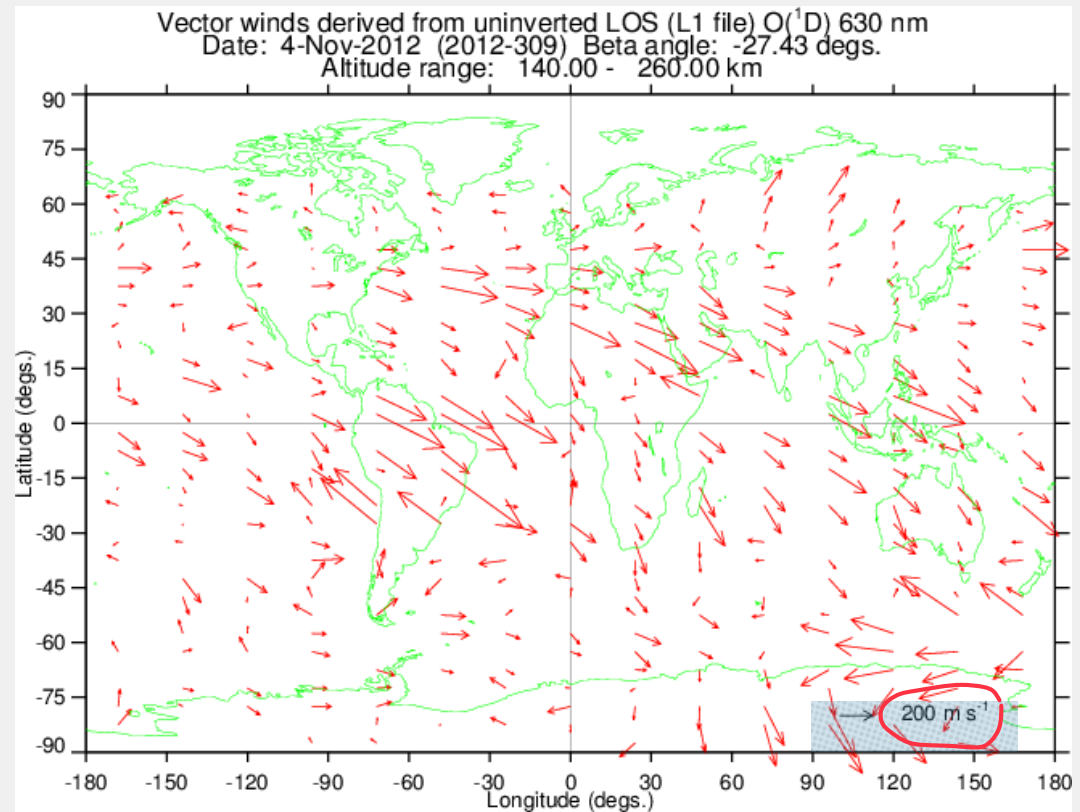
Winds

- ❖ Winds in the upper atmosphere are driven by at least four factors:
- ❖ Pressure gradient, day-to-night
- ❖ Coriolis force
- ❖ Viscosity
- ❖ Ion drag
- ❖ Different effects dominate in different regions of atmosphere (altitude, latitude)

- ❖ In Math: *Horizontal wind*

$$\frac{DU_h}{Dt} = -\frac{1}{\rho} \nabla_h p - \underbrace{2\Omega \times U_h}_{\text{coriolis}} + \underbrace{\frac{1}{\rho} \nabla(\mu \nabla U_h)}_{\text{viscosity term}} - \underbrace{\nu_{ni}(U_h - V_i)}_{\text{collisional ion drag}}$$

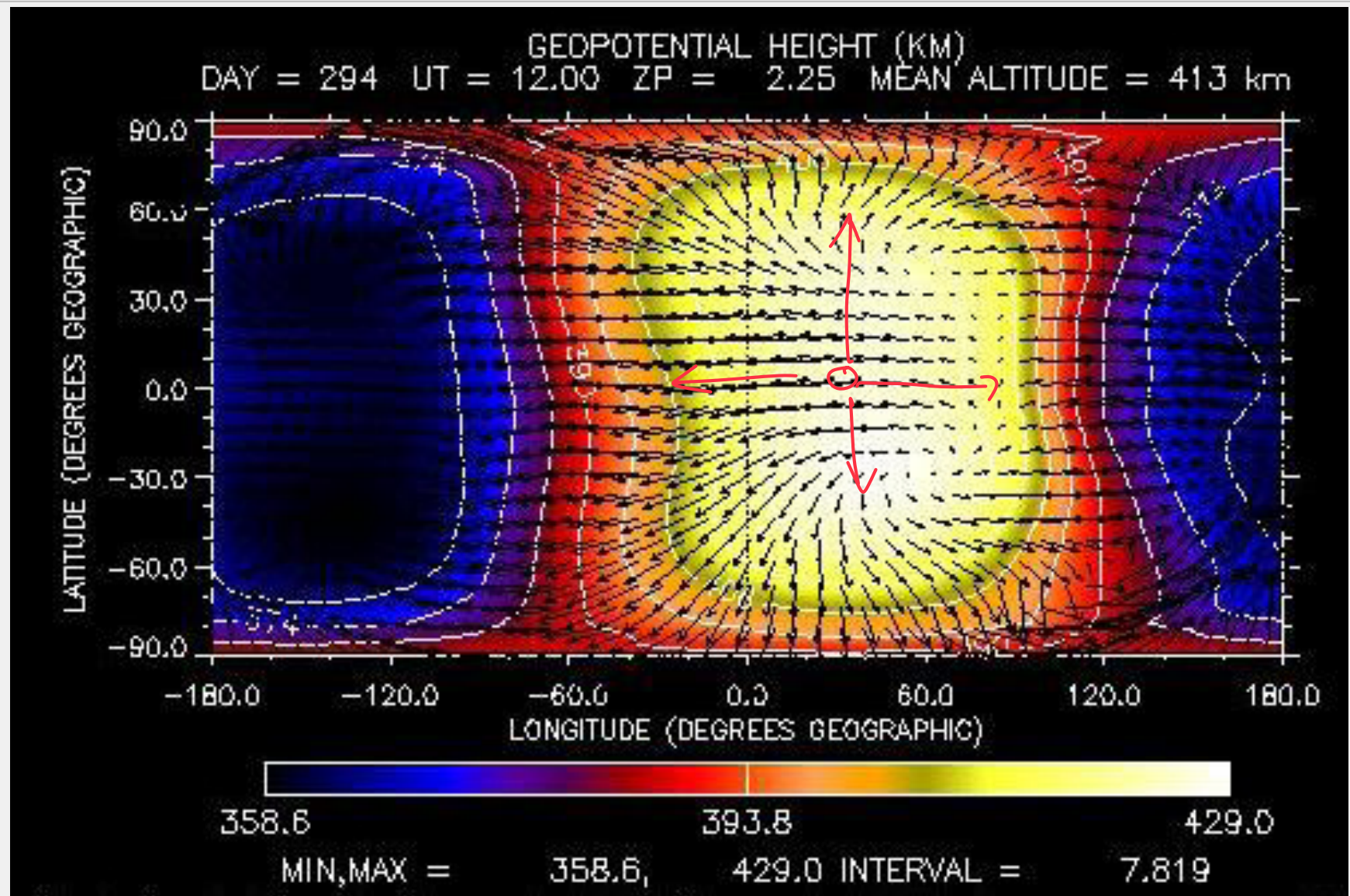
pressure gradient



from TIDI (TIMED Doppler Interferometer)
instrument on TIMED spacecraft

More looks at Thermospheric winds

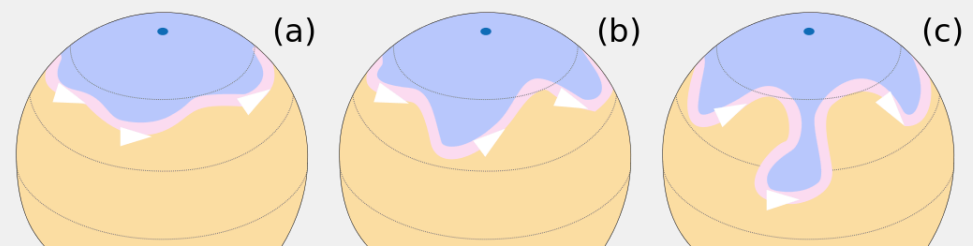
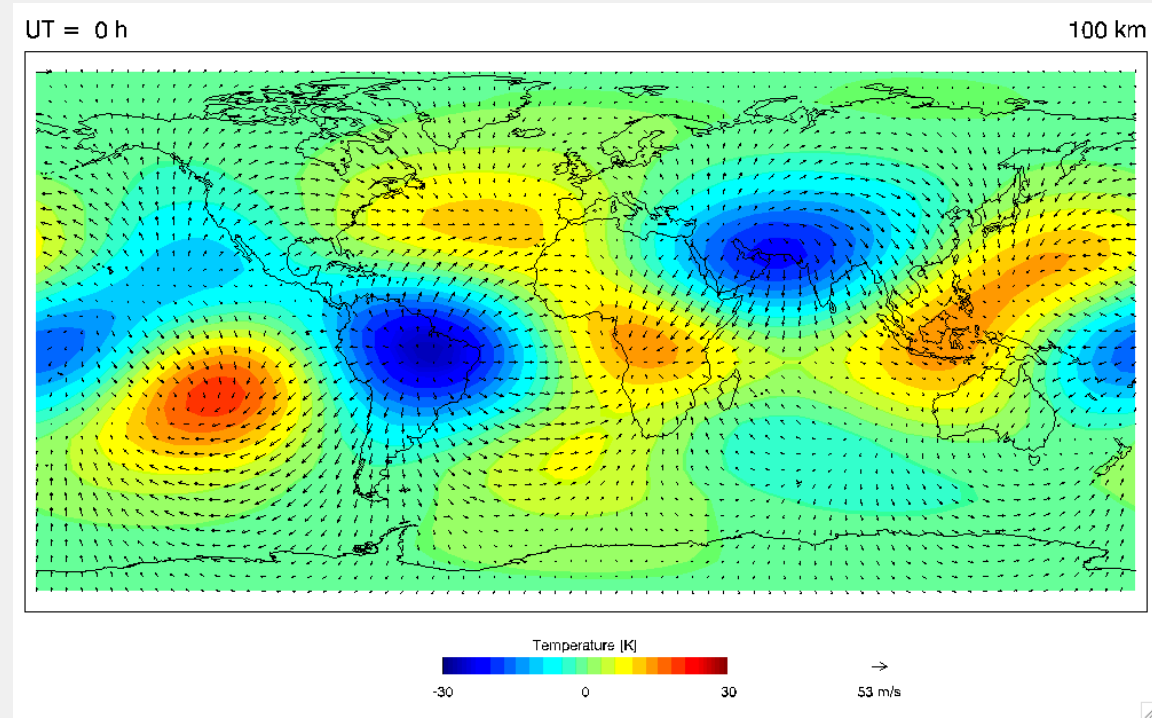
@ 413 km



pressure gradient dominated;
wind flows day to night

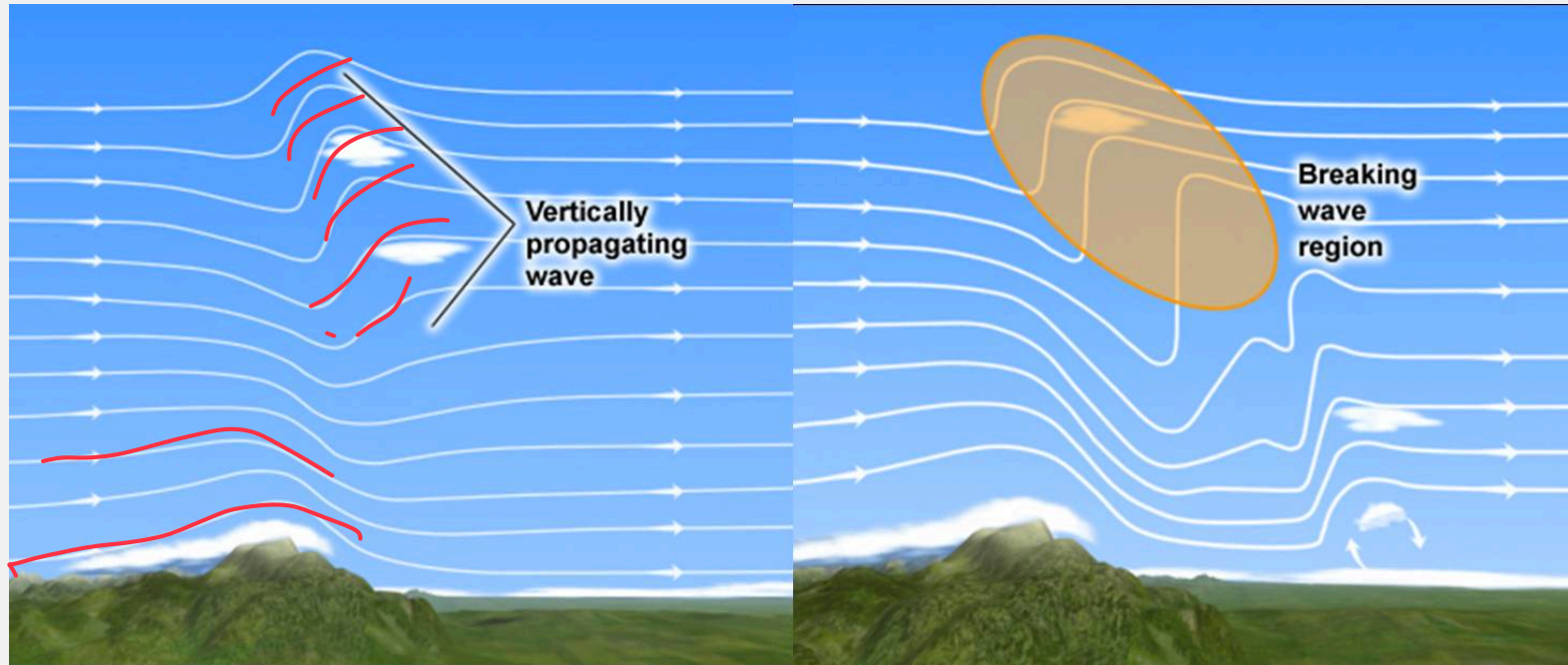
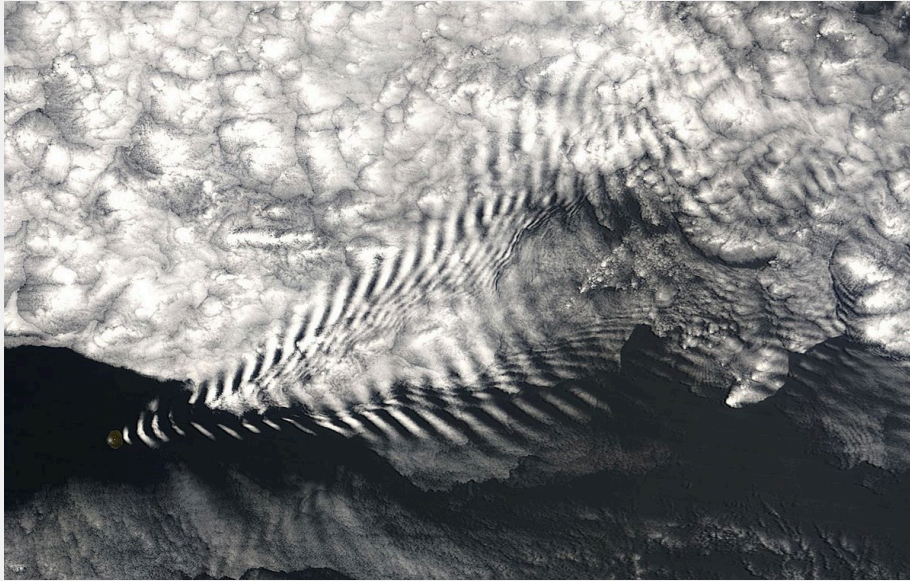
Tides

- ❖ Periodic oscillations in the atmosphere;
just like ocean tides
(but higher)
- ❖ **Caused by:**
 - ❖ Regular day-night cycle in the Sun's heating of the atmosphere
 - ❖ The gravitational pull of the Moon
 - ❖ Non-linear interactions between tides and planetary waves
 - ❖ Large-scale latent heat release due to deep convection in the tropics (thunderstorms)
- ❖ Can be migrating (sun-synchronous) or non-migrating
- ❖ Because atmosphere density falls with altitude, tide amplitude increases with altitude – exponentially!



Planetary (or Rossby) waves

Gravity Waves $\neq$ gravitational waves



Gravity Waves in upper atmosphere

- ❖ Similar to acoustic waves; derived from fundamental fluid equations
- ❖ Difference: **gravity (g) matters**. Allows waves to propagate with vertical direction
 - ❖ Bouyancy waves
- ❖ Amplitude is inversely proportional to pressure: grows exponentially with altitude
- ❖ Observed as temperature and/or density oscillations at particular altitudes

u, v, w pressure

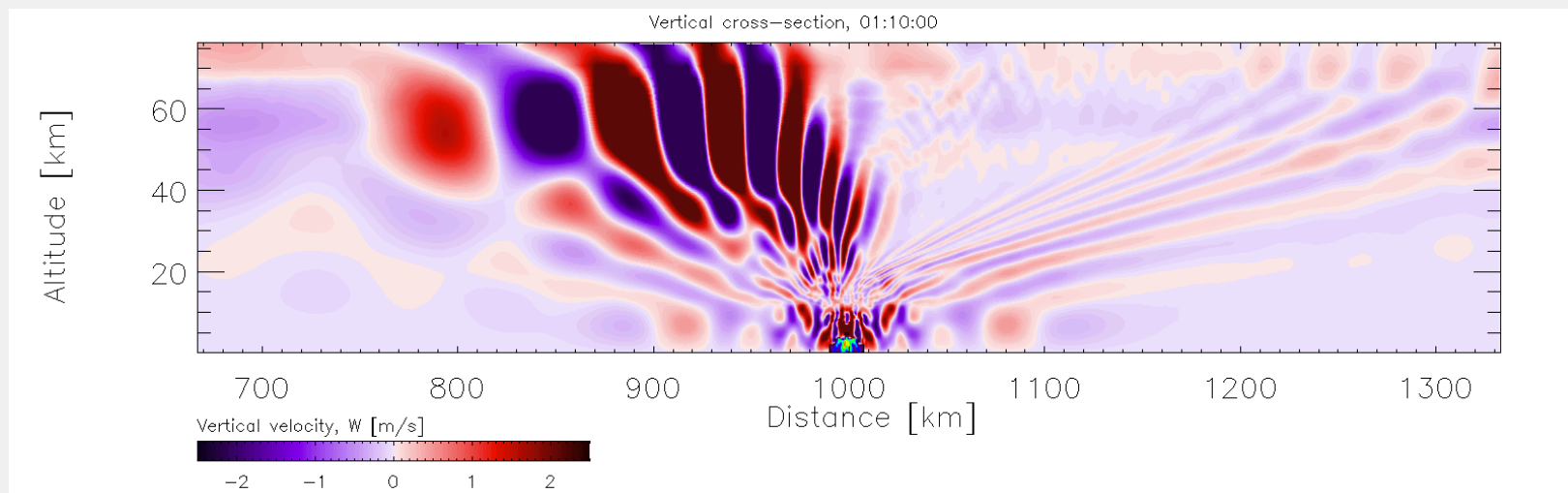
$$\frac{du}{dt} - fv + \frac{1}{\rho} \frac{\partial p}{\partial x} = X,$$

$$\frac{dv}{dt} + fu + \frac{1}{\rho} \frac{\partial p}{\partial y} = Y,$$

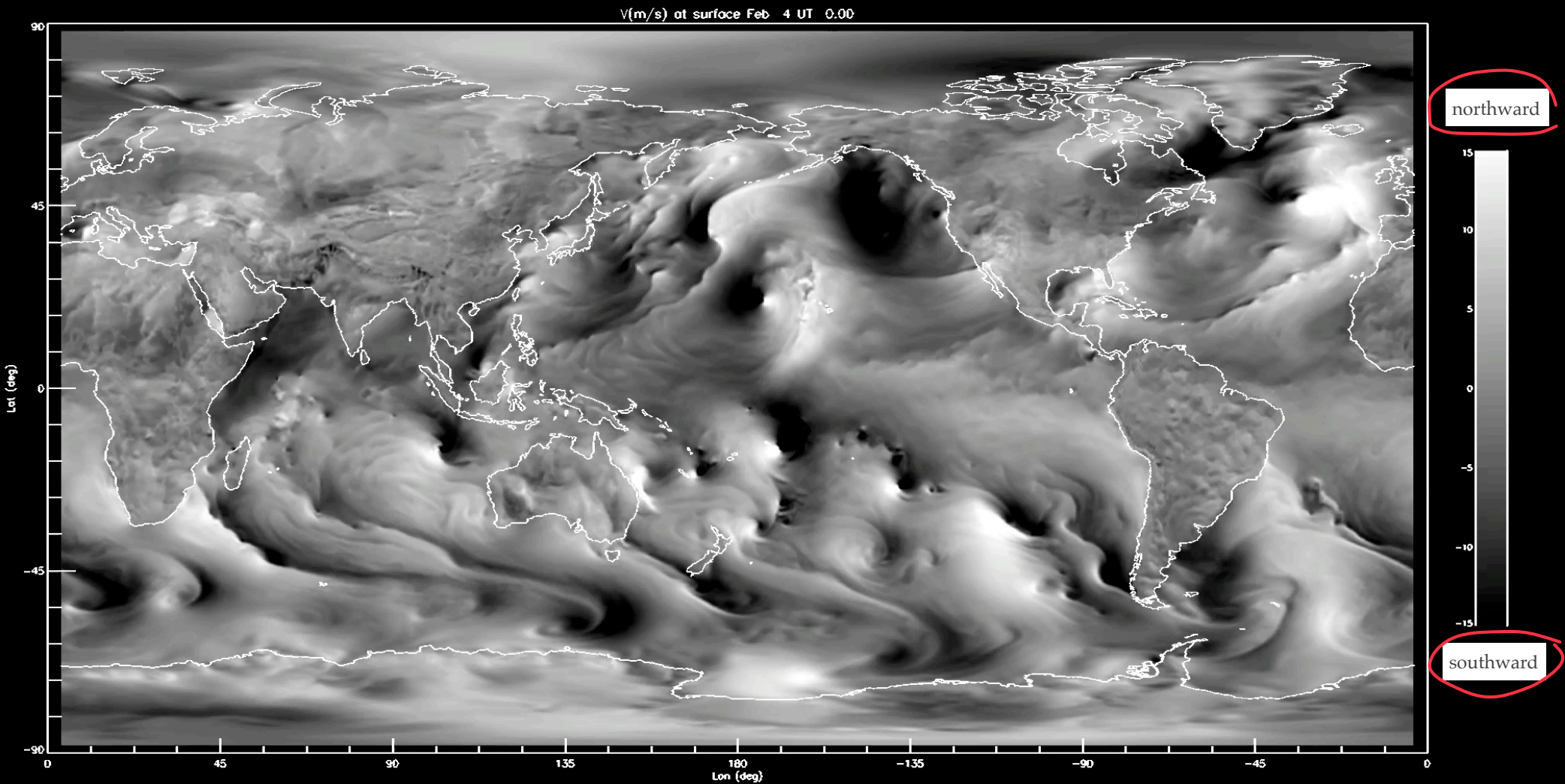
$$\frac{dw}{dt} + \frac{1}{\rho} \frac{\partial p}{\partial z} + g = 0,$$

$$\frac{1}{\rho} \frac{d\rho}{dt} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0,$$

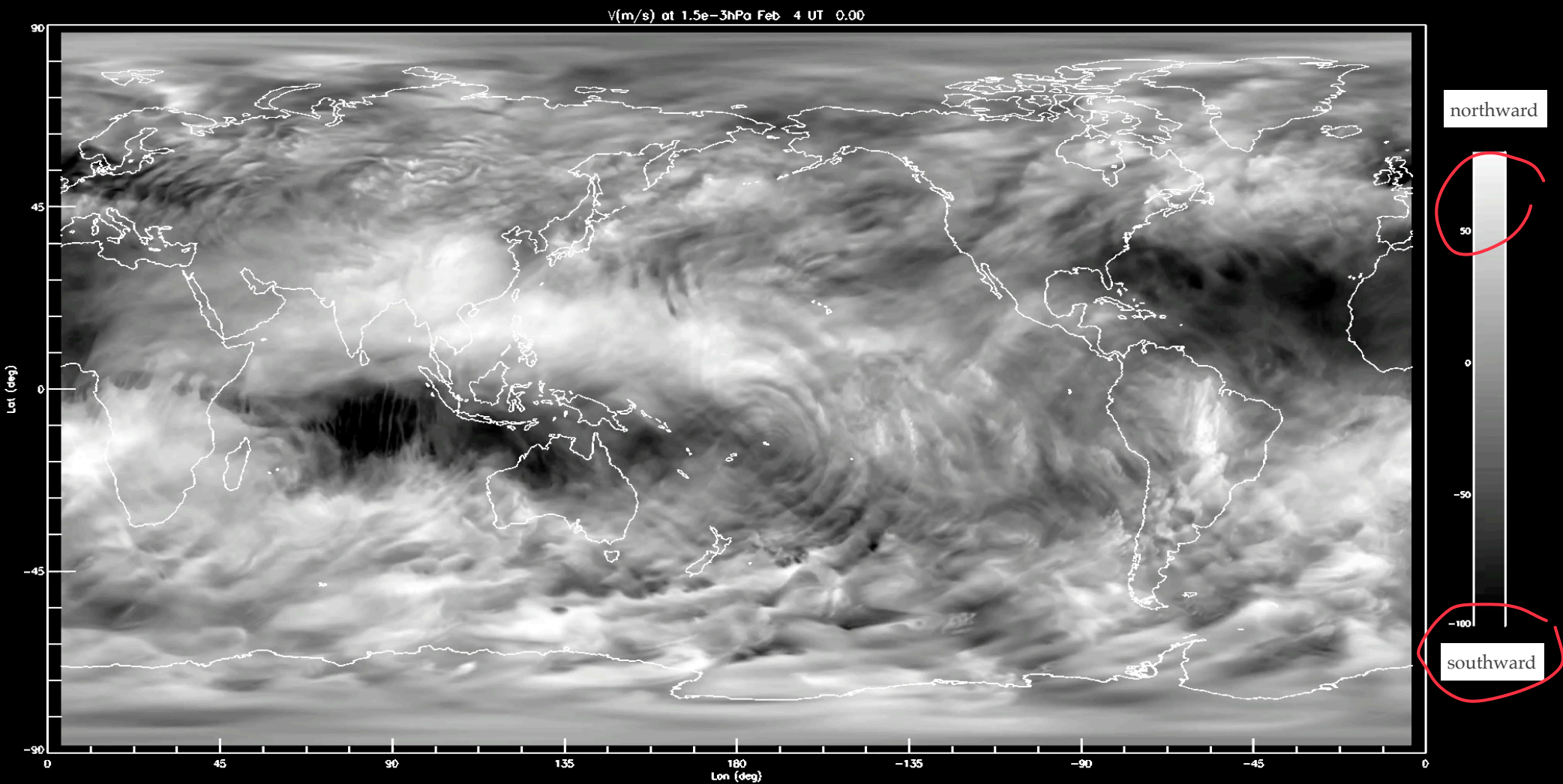
$$\frac{d\theta}{dt} = Q,$$



Meridional wind (m/s) from the same simulation, at Earth's surface (H. Liu et al., 2014)



Meridional wind (m/s) from the same simulation, at 100 km altitude (H. Liu et al., 2014)



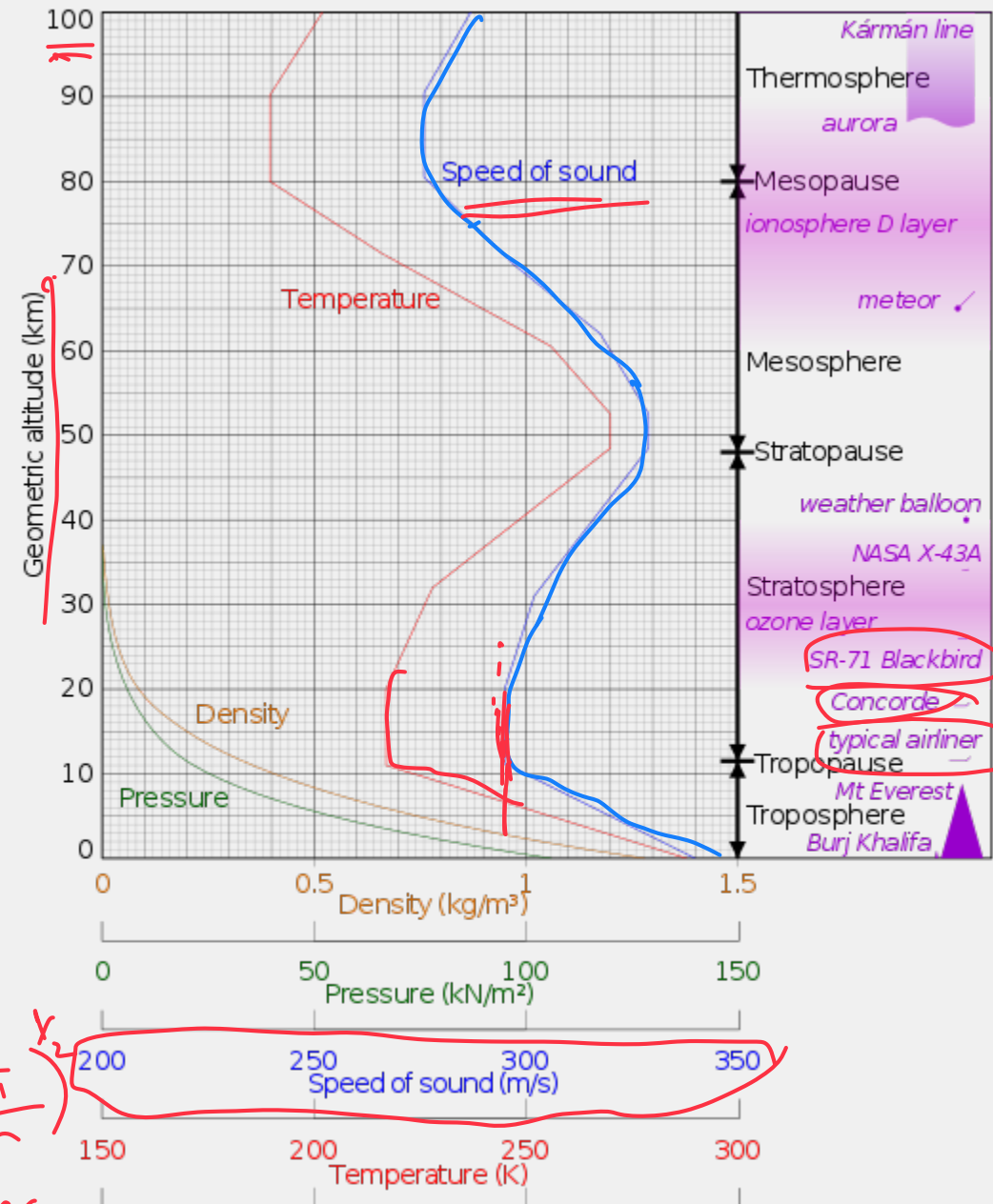
Why do all of these phenomena matter?

- ❖ **Science:** I just want to know how the atmosphere works.
Why? I just do.
- ❖ **Space Weather:** temperature changes, winds, waves, and tides all affect density and/or velocity of atmosphere, which affect spacecraft drag (as we'll see shortly)
 - ❖ as well as other effects on spacecraft and radio communications
- ❖ Winds / tides can also affect O/N₂ ratio, affecting drag and oxygen effects
- ❖ Gravity waves **break** at the bottom of the thermosphere; thought to cause **turbulence** that affects the height of the **turbopause**. But, as yet unverified!

Atmospheric Models

- ❖ MSIS: Mass Spectrometer and Incoherent Scatter radar (horrible name)
- ❖ MSIS-E-90: <https://www.ukssdc.ac.uk/wdcc1/msise90.html>
- ❖ NRLMSISE00: <https://ccmc.gsfc.nasa.gov/modelweb/models/nrlmsise00.php>
- ❖ Global, empirical models of atmosphere density, composition, and temperature
- ❖ Derived from data: Mass spec, radar, rocket measurements, and more
- ❖ Include variations with solar activity, location, and more
- ❖ Other models: US Standard Atmosphere, International Standard Atmosphere, Jacchia, others

US Standard Atmosphere

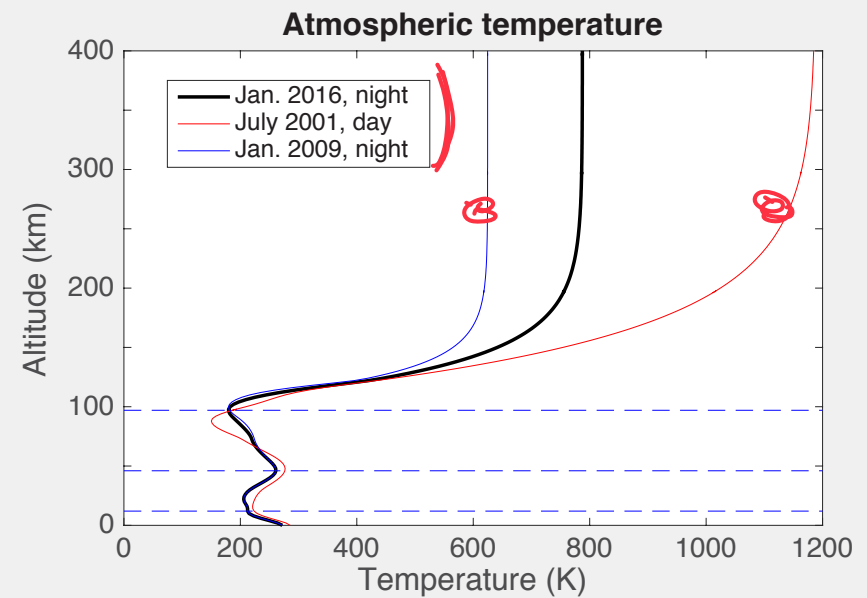
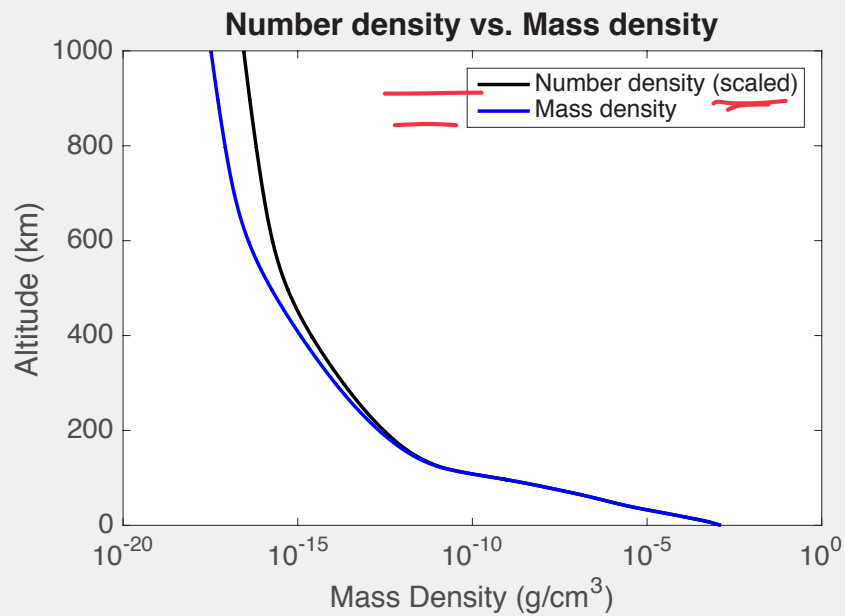
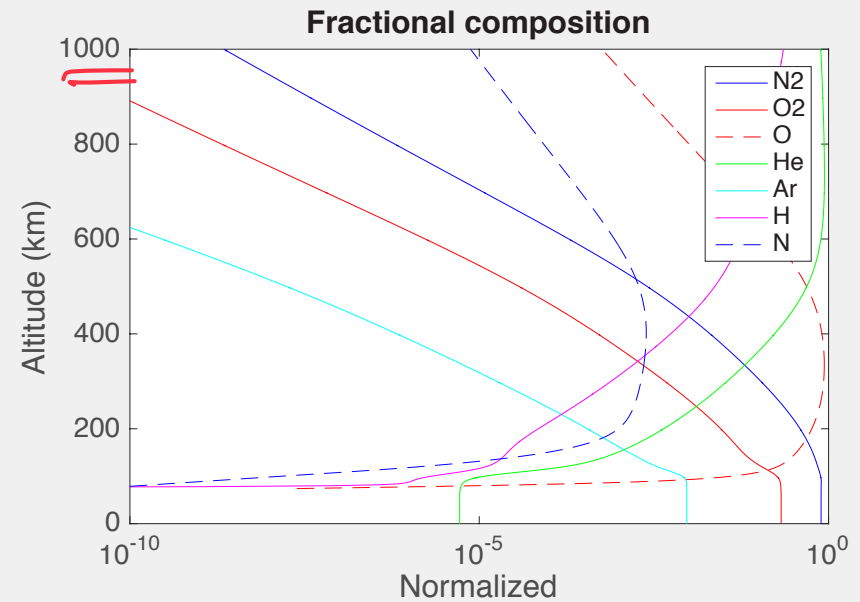
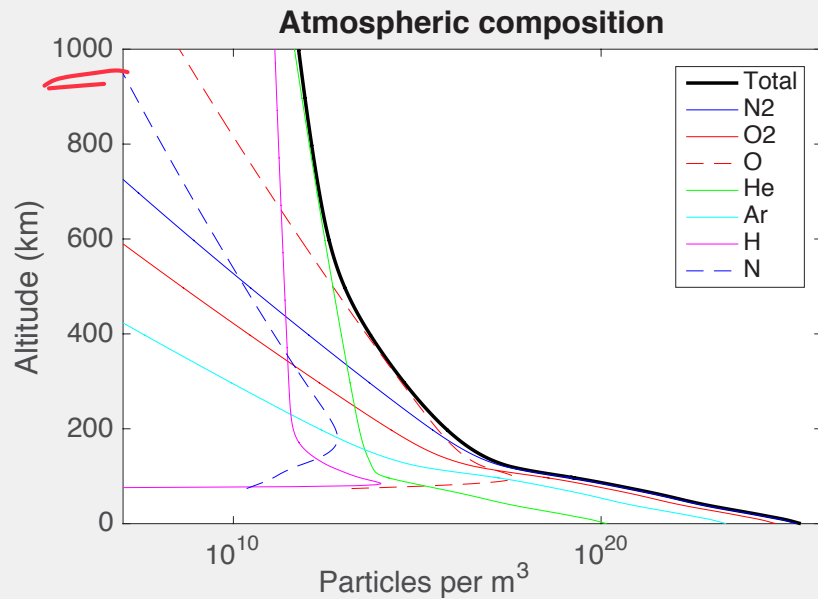


speed of sound:

$$c_s = \left(\frac{\gamma P}{\rho} \right)^{1/2} = \left(\frac{\gamma k T}{m} \right)^{1/2}$$

$\gamma = \frac{C_p}{C_v}$

Example output





University of Colorado **Boulder**

Atmospheric Effects

Currently Active Spacecraft

Satellite Quick Facts *(includes launches through 4/30/18)*

Total number of operating satellites: 1,886

United States: 859	Russia: 146	China: 250	Other: 631
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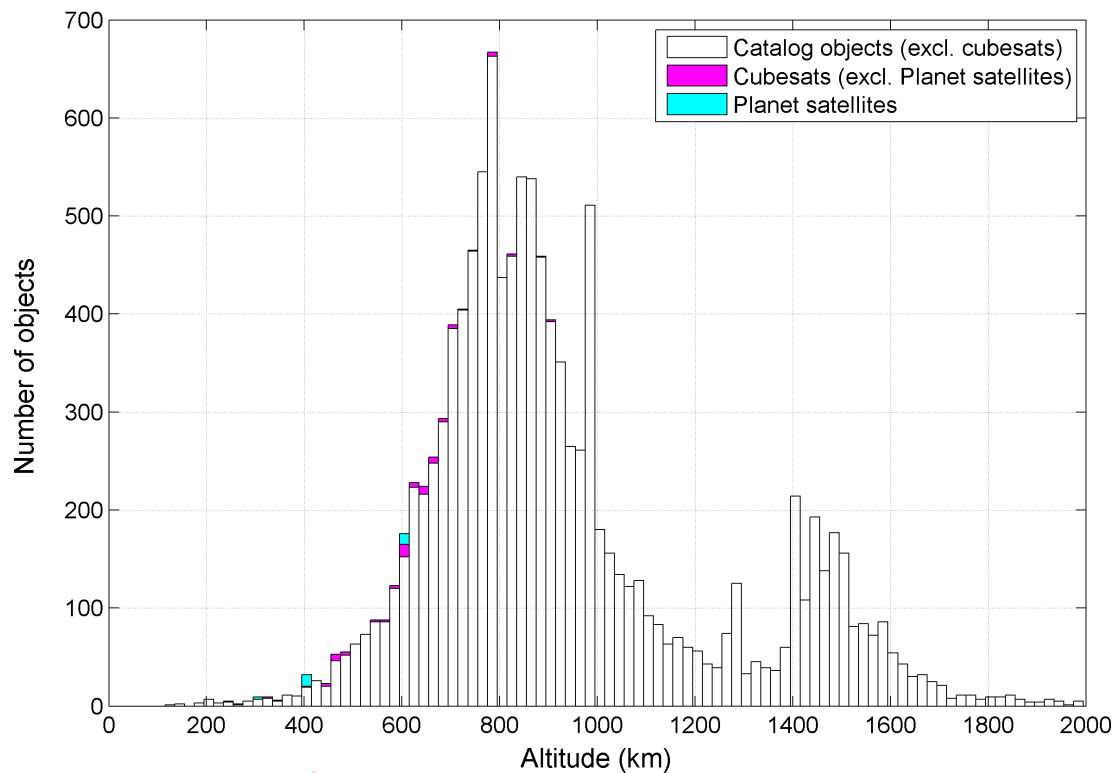
LEO: 1,186 <i>x20</i>	MEO: 112	Elliptical: 40	GEO: 548
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Total number of US satellites: 859

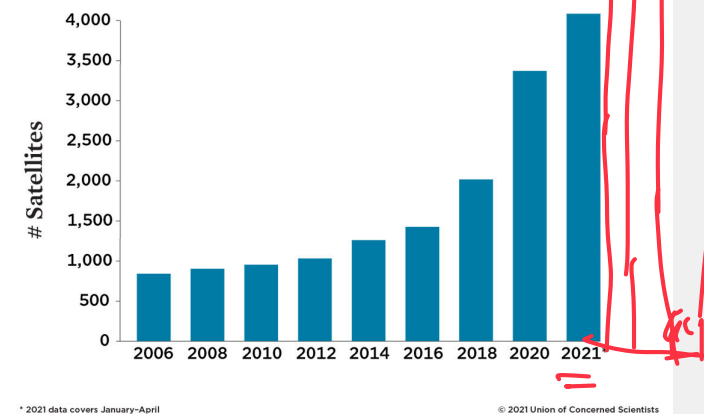
Civil: 20	Commercial: 495	Government: 178	Military: 166
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Most spacecraft are in LEO – subject to atmosphere and drag

Distribution of all objects including debris



Increase in Operating Satellites



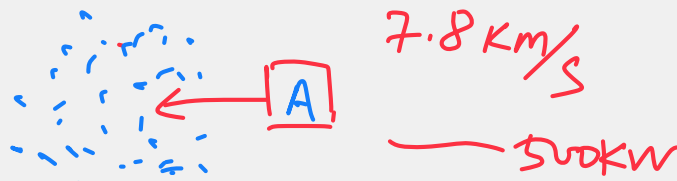
Satellite Drag

- ❖ The atmosphere imposes a drag force on any object (just as you'd expect)

$$\Delta P \text{ (momentum)} = P_i + \underline{P_r}$$

$$= \underline{P_i} \left(1 + \frac{P_r}{P_i} \right) = P_i \left(1 + f(\theta) \right)$$

$$\Delta P = \rho A V_i^2 \left(1 + f(\theta) \right) \Delta t = \underline{m} V_i \left(1 + f(\theta) \right)$$



 7.8 km/s
 500 km
 A: area

$m = \rho A V_i \Delta t$
 ΔVolume

drag force:

$$F_D = \frac{\Delta P}{\Delta t} = \rho A V_i^2 \left(1 + f(\theta) \right)$$

$$F_D = \frac{1}{2} \rho V_i^2 A C_D$$

$C_D = 2(1 + f(\theta))$: drag coeff.

$$C_D = \frac{F_D/A}{\frac{1}{2} \rho V_i^2} = \frac{\text{Pressure}}{\text{dynamic pressure}}$$

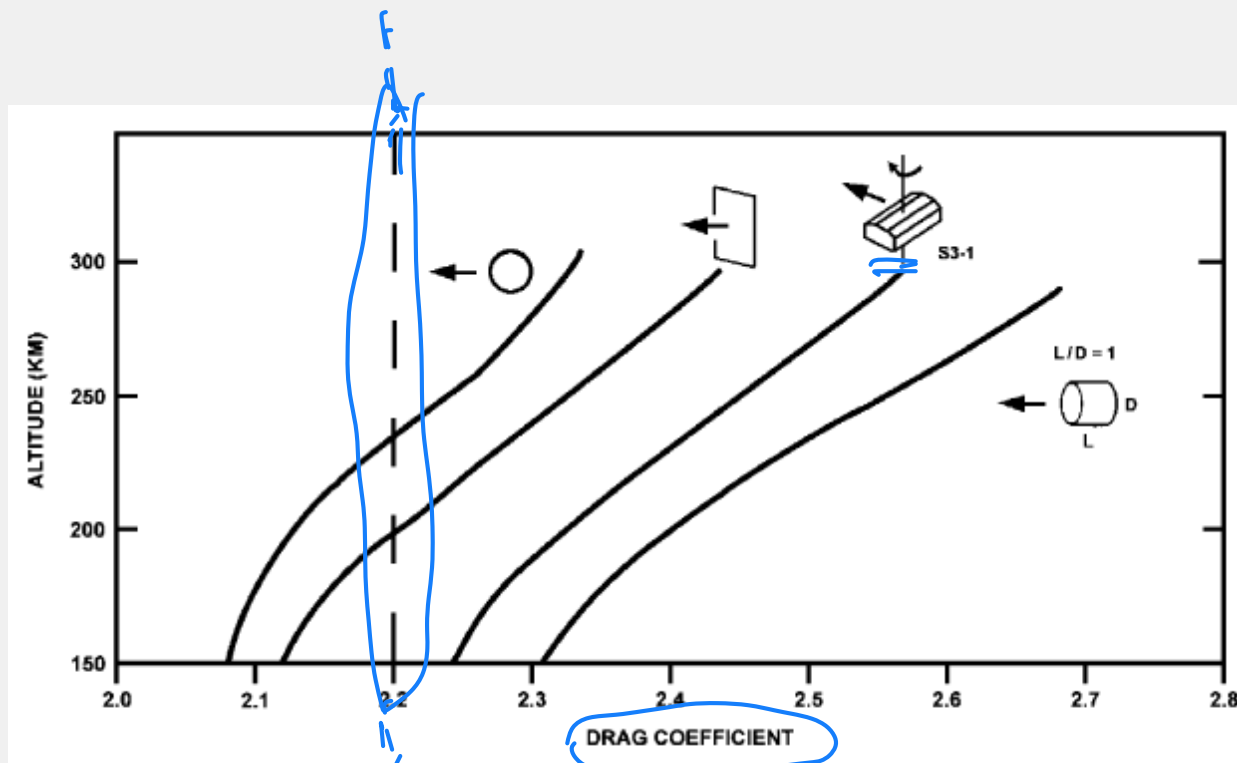
well defined
 well determined

most uncertain!!
orders of magnitude

uncertain
 a factor "2" or 20

Drag coefficient

- ❖ Drag coefficient is particularly hard to measure!
- ❖ Lower-Left: drag coefficients in continuous flow regime
- ❖ In space: rarefied (free molecular) flow!



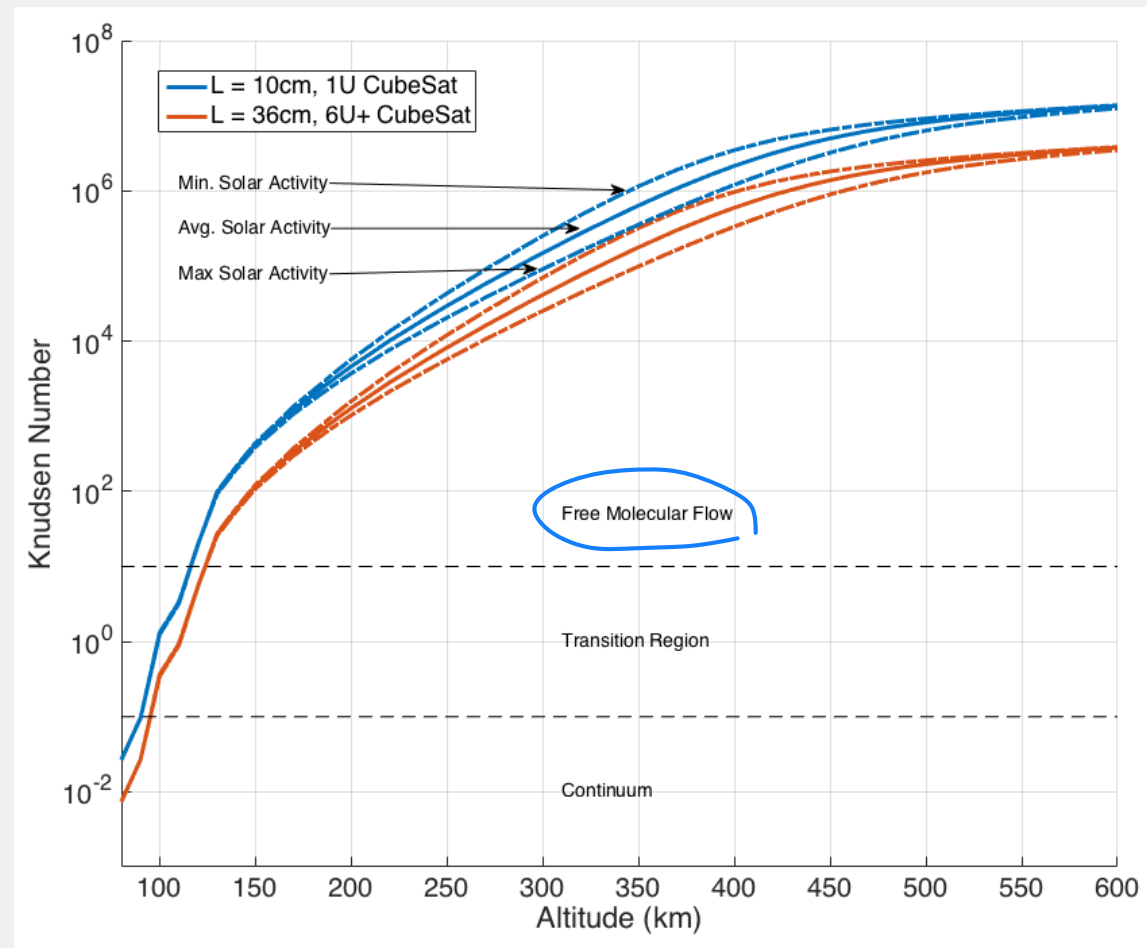
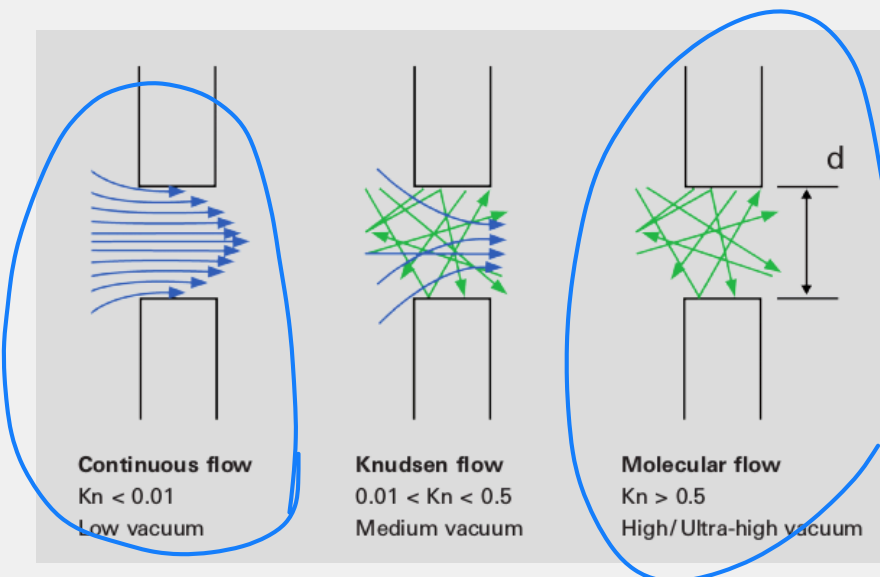
Shape	Drag Coefficient
Sphere	0.47
Halfsphere	0.42
Cone	0.50
Cube	1.05
Angled Cube	0.80
Long Cylinder	0.82
Short Cylinder	1.15
Streamlined Body	0.04
Streamlined Halfbody	0.09

Measured Drag Coefficients

in lab

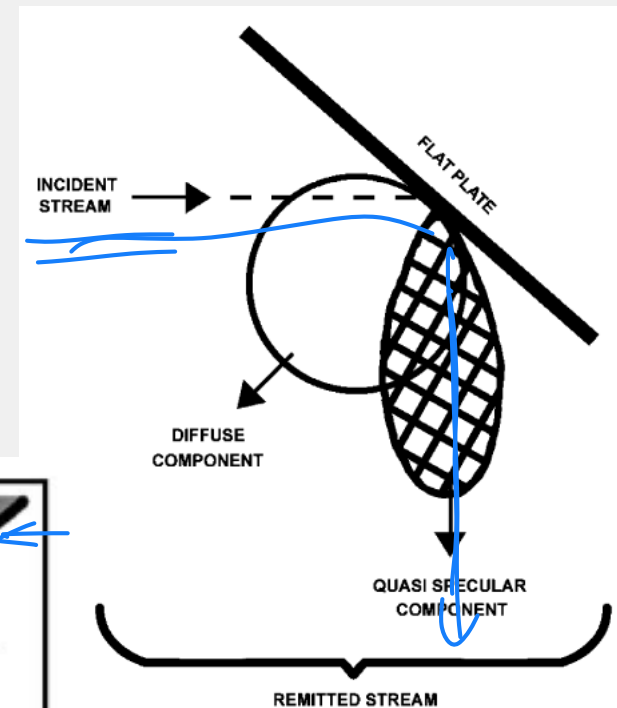
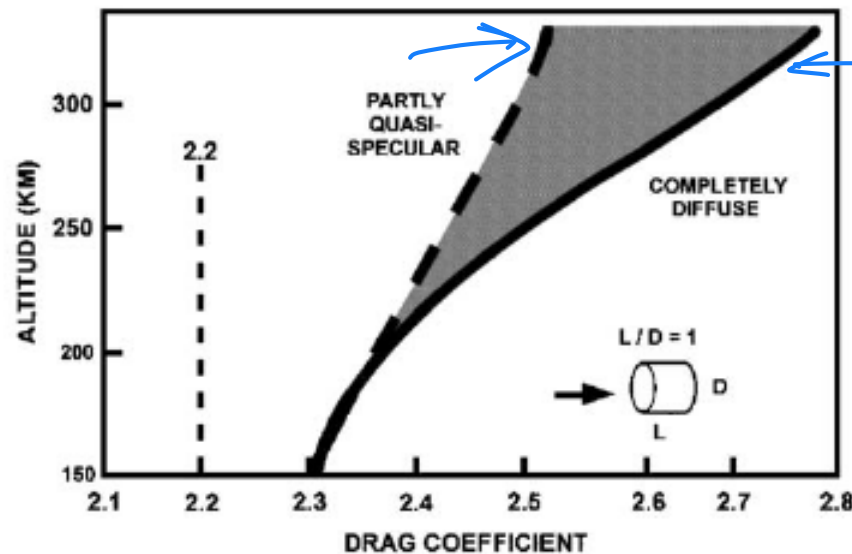
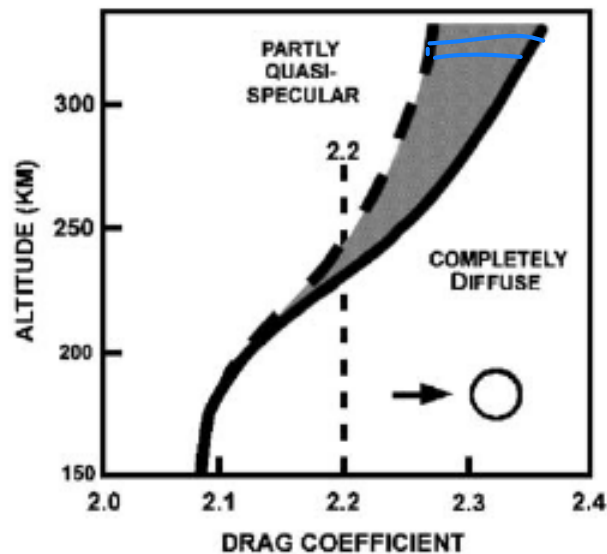
Continuous vs. Free Molecular Flow

- ❖ Knudsen number: $Kn = \lambda/L$
- ❖ λ = **mean free path** (m)
between collisions
- ❖ L = length scale (object)
- ❖ Basically, are we in a “flow”,
or just getting hit by
molecules / atoms one at a time?



Drag coefficient and Scattering

- ❖ Drag coefficient depends on how molecules scatter from surfaces: either **specular** (mirror reflection) or **diffuse** (scattering from an opaque surface)
- ❖ This scattering is related to **accommodation**, which measures scattering elasticity
 - ❖ accommodation coefficient (≤ 1):



Does Drag make satellite speed up, or slow down?

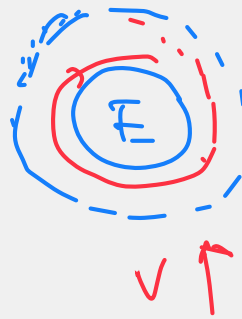
- ❖ Kepler's third law relates period to altitude
(for circular orbit, semi-major axis $a = r$)

circular orbit

$$P^2 = \frac{4\pi^2}{GM} r^3$$

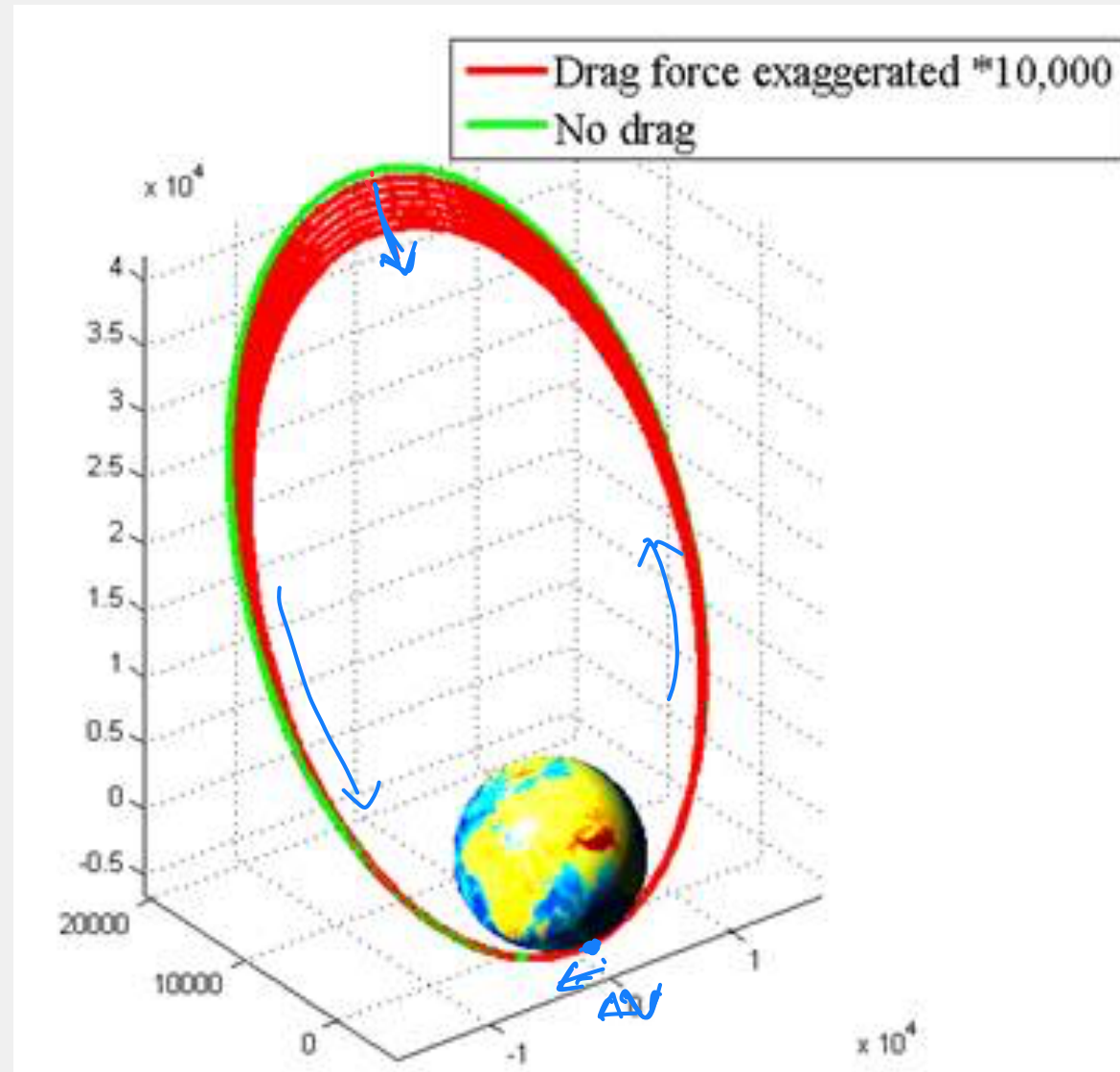
$$v = \sqrt{\frac{GM}{r}}$$

$$r \downarrow \Rightarrow v \uparrow$$



Drag effect on elliptical orbit

- ❖ Drag force slows spacecraft down; now it can't reach as high of an apogee
- ❖ Returns to same perigee
- ❖ After a long time, orbit will be circular, with altitude = original perigee



Magnitude of Drag Force

$$h = 400 \text{ km} \quad \rho = 10^{-15} \text{ g/cm}^3$$

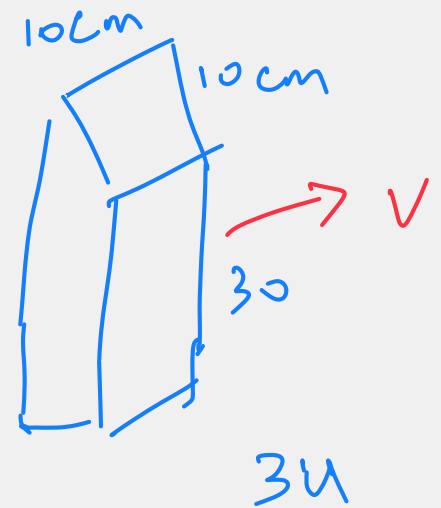
$$V = 8 \text{ km/s}$$

$$A = 0.1 \times 0.3 = 0.03 \text{ m}^2$$

$$C_D = 2.2$$

$$F_D = \frac{1}{2} \rho v^2 C_D A \sim \underline{\underline{2 \text{ MN}}}$$

$$F_g = 4 \text{ kg} \times \underline{\underline{9.5 \text{ m/s}^2}} \sim \underline{\underline{38 \text{ N}}}$$

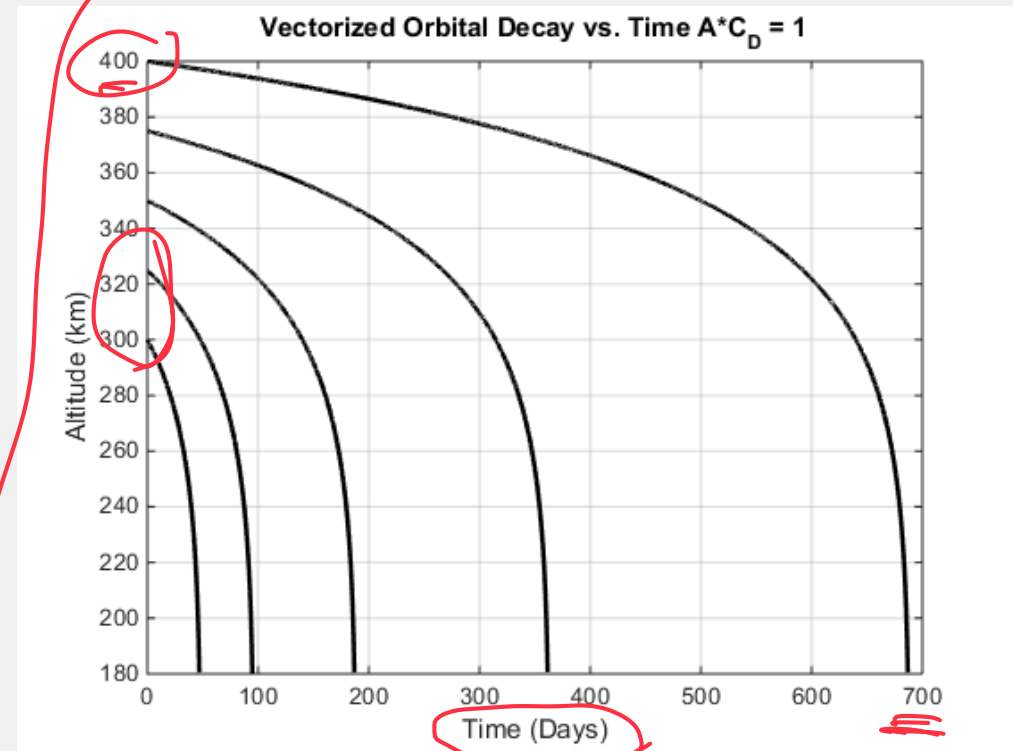


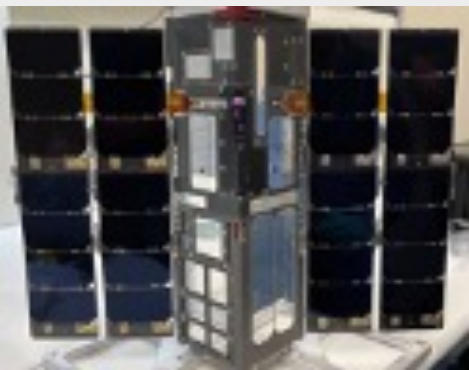
Orbit Degradation

- ❖ Differentiate Kepler's third law to determine the change in period with time (HW)
- ❖ period decreases with higher drag, higher area, density, and altitude
 - ❖ but density decreases with altitude
- ❖ Spacecraft mass (m) reduces change in period; "ballistic"
- ❖ From dP/dt , can also determine change in altitude; how long until it reaches some minimum altitude (e.g., 120km)?

$$\frac{dP}{dt} \rightarrow \frac{\Delta P}{\Delta t} = \frac{P_{i+1} - P_i}{\Delta t}$$

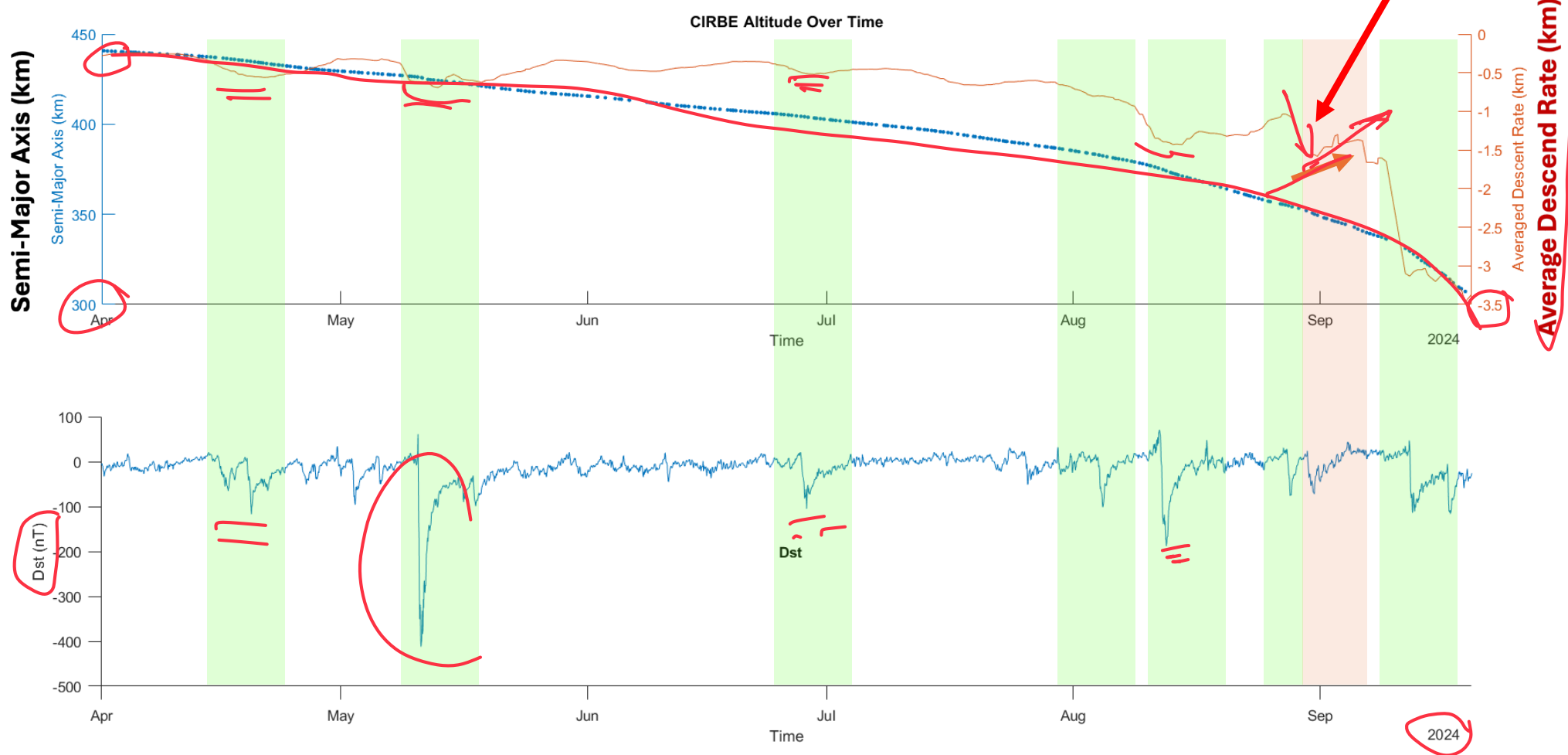
$$\frac{dP}{dt} = -3\pi r \rho \frac{C_D A}{m}$$





CIRBE's orbit altitude since April of 2024

Low Air-Drag
Flying Mode



ISS altitude vs time

